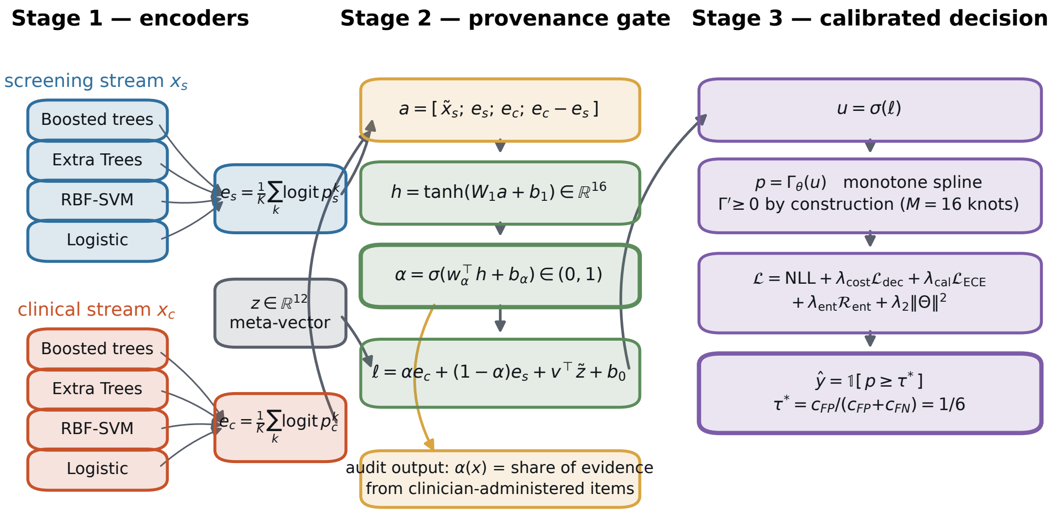
Stage 1 — encoders
Stage 2 — provenance gate
Stage 3 — calibrated decision
screening stream x_s
Boosted trees
Extra Trees
RBF-SVM
Logistic
$e_s = \frac{1}{K}\sum_k \operatorname{logit} p_s^k$
clinical stream x_c
Boosted trees
Extra Trees
RBF-SVM
Logistic
$e_c = \frac{1}{K}\sum_k \operatorname{logit} p_c^k$
$z \in \mathbb{R}^{12}$ meta-vector
$a = [\,\tilde{x}_s;\ e_s;\ e_c;\ e_c - e_s\,]$
$h = \tanh(W_1 a + b_1) \in \mathbb{R}^{16}$
$\alpha = \sigma(w_\alpha^\top h + b_\alpha) \in (0,1)$
$\ell = \alpha e_c + (1-\alpha) e_s + v^\top \tilde{z} + b_0$
audit output: $\alpha(x)$ = share of evidence from clinician-administered items
$u = \sigma(\ell)$
$p = \Gamma_\theta(u)$ monotone spline
$\Gamma' \geq 0$ by construction ($M = 16$ knots)
$\mathcal{L} = \mathrm{NLL} + \lambda_{\mathrm{cost}}\mathcal{L}_{\mathrm{dec}} + \lambda_{\mathrm{cal}}\mathcal{L}_{\mathrm{ECE}} + \lambda_{\mathrm{ent}}\mathcal{R}_{\mathrm{ent}} + \lambda_2\|\Theta\|^2$
$\hat{y} = \mathbb{1}[\,p \geq \tau^*\,]$
$\tau^* = c_{FP}/(c_{FP}+c_{FN}) = 1/6$